

A working list of Australian birds

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This document outlines the development and management of a working list of Australian birds (hereafter referred to as *wlist*). The document serves two purposes: (1) to describe the conceptual basis of the working list, the ontology of which is described in sections 1-3, and (2) to describe technical details which are covered in sections 4-6.

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1. Introduction

1.1. **Origins**

This working list of Australian birds (*wlist*) has been developed as part of a broader effort to support conservation planning, ecological research, and biodiversity data management in Australia. It draws on a wide range of foundational resources, in particular the Action Plans for Australian Birds (Garnett and Crowley 2000, Garnett et al. 2011, Garnett and Baker 2021), which have, as a matter of course, utilised the taxonomic works of the day as a basis for classifications (e.g. Christidis and Boles 1994, Schodde and Mason 1997, Schodde and Mason 1999, Christidis and Boles 2008).

Since its inception (Ehmke 2013), the species classifications have been progressively aligned with global avian taxonomies, initially the BirdLife International checklist (versions 5-9, 2012-2017), subsequently HBW/BirdLife Checklists (<https://datazone.birdlife.org/about-our-science/taxonomy>) and most recently through integration with the AvIList Core Team (see Section 3 and Appendix 1).

Subspecies listings have been drawn from the seminal works of Schodde and Mason (1997), Schodde and Mason (1999), the Handbook of Australian, New Zealand and Antarctic Birds (Marchant and Higgins 1990), later the HBW and BirdLife International Illustrated Checklist of the Birds of the World (del Hoyo et al. 2014, del Hoyo et al. 2016) and published taxonomic works.

The list has been developed in parallel with a suite of related datasets, including spatial range layers, ecological trait databases, conservation assessments, and bird occurrence records. It provides a shared taxonomic framework linking these resources.

Many have contributed to this list, notably the BirdLife Australia, English Names Committee and the BirdLife Australia Threatened Species Committee, along with many experts who kindly lent their expertise and criticism.

1.2. **Purpose**

The *wlist* is a **working inventory of Australian birds** designed to support applied conservation and ecological analysis. Its primary purpose is to provide a stable and consistent framework for organising biodiversity data while accommodating ongoing changes in taxonomy.

Taxonomy is treated as an evolving scientific framework, whereas ecological observations and conservation data must remain interpretable through time. Accordingly, this list separates **persistent biological observations** from **changing taxonomic interpretations** (see Section 2). This allows datasets to remain internally consistent even as classifications are revised. To achieve this, the list:

- maintains stable, unique identifiers for all taxa (see Section 5, Table 2),
- records taxa at the finest practical resolution (see Section 4), and
- aligns with global taxonomies while allowing transparent, limited deviations (see Section 3).

This approach supports long-term data usability, reproducibility of analyses, and integration across diverse datasets.

***Key principle:** Observations are stable; taxonomy changes. The list preserves the former while tracking the latter. The *wlist* treats the rapidly advancing molecular science that increasingly underpins evolutionary biology as an evolving and critically important layer of knowledge, applied to persistent observational data, providing a basis for preserving the finest diagnosable biological units recognised at the point of data collection and ensuring that conservation analyses remain valid under future taxonomic changes.*

1.2.1. Role in extinction risk status and data systems

Taxonomy underpins conservation assessment, legislative protection, and ecological research (Mace 2004, Garnett and Christidis 2007, Burfield et al. 2017). Global Red List statuses are maintained by the International Union for the Conservation of Nature (IUCN). However, the IUCN only assesses species, whereas in Australia, many highly imperilled birds are subspecific.

***Note:** Conservation status is the legal standing of a taxon under legislation (Garnett et al. 2019). Conservation status listings (e.g. EPBC status) are maintained by government agencies and are not included in this list.*

Comprehensive listings of all bird species and subspecies in Australia were pioneered, by necessity, by the Action Plans for Australian Birds. Australian Red List Status has been systematically assessed using IUCN criteria (IUCN 2025) on a decadal basis (Garnett 1992, Garnett and Crowley 2000, Garnett et al. 2011, Garnett and Baker 2021). However, intervening updates have been regularly conducted by the BirdLife Australia Threatened Species Committee, updating the published assessments in successive Bird Action Plans. These updates lacked a formal mechanism for publication. Maintaining an up-to-date inventory of extinction risk status was a driver behind the formation of this resource.

***Key principle:** Current, comprehensive listings of extinction risk are a critical resource for conservation planning and research.*

The *wlist* provides a technical foundation for linking and integrating multiple data resources, including:

- spatial range datasets,
- ecological and life-history trait databases,
- conservation status assessments, and
- bird occurrence records.

By providing a consistent taxonomic reference and stable identifiers, the list enables these datasets to be combined and analysed in a coherent and reproducible way.

1.2.2. The strategic role of vernacular nomenclature

While scientific names are the basis of scientific classification, common (vernacular) names are vital for communication, data management, and public engagement. Vernacular names often reflect key characteristics helpful to observers and they have remained relatively stable compared to their scientific counterparts (Ehmke *et al.* 2017).

The *wlist* systematically maintains vernacular names in accordance with Australian naming principles (Royal Australasian Ornithologists Union 1978) and subsequent practice through field guides (e.g. Menkhorst *et al.* 2017) and provides comprehensive vernacular names for core subspecies following the principles set out in (Ehmke *et al.* 2017).

Common names are treated as structured data fields rather than informal labels, supporting their use in databases, reporting, and communication (see Section 5). This includes the standardisation of subspecies names where appropriate.

***Example:** The devil is in the detail - Though increasing the length of names, the word 'Island' is included for island endemic birds following (Garnett 1992) to avoid the names that imply birds are characteristic of late December ('Christmas' taxa), are of East Anglia ('Norfolk' taxa), are regal ('King' taxa), are a strange mix of mammal and bird ('Kangaroo' taxa) etc. The word 'island' is omitted in a few cases for brevity where the island name is not conflicted, or where existing precedents are in place - e.g. for Lord Howe Island endemics because of the established protocol for the high-profile 'Lord Howe Woodhen'.*

Subsequent sections describe the challenges posed by taxonomic change (Section 2), the scope and alignment of the list (Section 3), the biological units it recognises (Section 4), and the technical structure that underpins its use (Section 5).

2. Why changing taxonomy is a problem for conservation

2.1. **Instability and competing concepts**

Biological taxonomy provides the fundamental framework for understanding and conserving biodiversity. In practice, a stable system for classifying and naming birds underpins data collection, legislative protection, and the interpretation of evolutionary relationships. However, taxonomy is inherently dynamic. Because biological diversity is continuous rather than discrete, the delineation of species and subspecies is partly interpretive, and multiple competing species concepts are in use. As a result, classifications change over time as new evidence emerges. This instability reflects scientific progress and the accumulation of knowledge. However, it creates practical challenges for conservation and data management.

Data are collected at a moment in time, but the taxonomic interpretation of those observations may change. The same organism, observed at the same location, may be classified differently across successive taxonomies—for example, as a species, then a subspecies, or as part of a different taxon altogether.

***Example:** The Christmas Island Goshawk, for instance, has been variously classified as a distinct species, a subspecies of Brown Goshawk, a subspecies of Variable Goshawk, and subspecies of Grey Goshawk.*

In some cases, taxonomic revisions alter the resolution at which biodiversity is recorded. Changes such as the merging of taxa ("lumping") can result in the loss of biologically meaningful units, particularly where distinct populations are no longer formally recognised. This is especially relevant for subspecies and other terminal units used in conservation (see Section 4).

***Key principle:** If taxonomy changes but data are not structured to accommodate those changes, analyses may become inconsistent or misleading through time.*

Given these challenges, stable data structures are required to ensure that biological observations remain interpretable through time, regardless of changes in classification (see Section 5).

2.2. **Conservation consequences**

Conservation value is not inherently determined by taxonomic rank. In Australia, legislation and conservation practice recognise taxa at the species, subspecies, and population levels. In Australia, multiple species concepts have been applied to bird lists over time, contributing to substantial taxonomic flux. The consequences for conservation and data management can be significant, and at times have been described as "anarchic" (Garnett and Christidis 2017).

Revisions such as species "splits" (one species reclassified as several) and "lumps" (multiple species merged) can disrupt long-term datasets, introduce apparent gaps or discontinuities, and necessitate extensive updates to legislation and key data resources. These changes may not always yield proportional conservation benefits, yet they can erode the stability of monitoring frameworks and weaken the institutional and social capital that effective conservation depends on.

***Example:** Taxonomic uncertainty need not be conservation uncertainty. The Western Ground Parrot is a prime example of the high-level uncertainty in taxonomic classification. Having been variously classified as a subspecies (1792), then a species (1911), then a subspecies (1912), then a species (2010), and then a subspecies again (2012), the critical conservation status has remained unchanged. It is one of our most imperilled birds.*

To address this, the working list adopts a system in which observations are anchored to stable taxonomic identifiers, while taxonomy is treated as an evolving interpretive layer (see Section 5: *How the list works*). This allows datasets to remain internally consistent while accommodating updates from global taxonomies such as Avilist Core Team (see Section 3 and Appendix 1).

2.3. **Taxonomic equality**

Despite assumptions, a bird need not be classified as a species in order to be valued. In Australia, conservation legislation at both state and federal levels enables the protection of birds at the species, subspecies, and even population levels. Many conservation-dependent birds are subspecific taxa (Figure 1), and their recognition is essential for effective management. Accordingly, this list treats all terminal taxonomic units (ultrataxa; see Section 4) as equally valid for the purposes of data management and conservation. Taxonomic classification is used to organise biological knowledge, but does not confer priority or importance in itself.

In summary...

Taxonomy is an evolving scientific framework, whereas conservation and ecological data require stability through time. The central challenge is therefore to accommodate taxonomic change without compromising data integrity or conservation effectiveness.

This working list addresses that challenge by:

- anchoring observations to stable identifiers (Section 5),
- maintaining the finest practical biological units (Section 4), and
- aligning with and complementing, but not being constrained by, global taxonomies (Section 3; Appendix 1).

Together, these elements ensure that data remain usable, comparable, and interpretable over time, despite ongoing changes in classification.

3. Scope

3.1. **A global basis for national working data**

While endemic birds are exclusively the domain of Australia, many birds occur across political boundaries. Many migratory species feature in bilateral or multilateral treaties and agreements aimed at their conservation. Aligning with global frameworks, therefore, supports comparability in conservation assessments, monitoring programs, and international reporting. This has been notoriously difficult given the proliferation of global bird checklists over the last 20 years (Garnett and Christidis 2017), however the inaugural release of AviList brings to fruition the alignment of global bird lists.

The list now follows the taxonomy of AviList (AviList Core Team 2025) as its primary global reference. Vernacular nomenclature follows Australian naming principles (see Section 1; Purpose). This provides a consistent and contemporary baseline for species-level classification, while enabling integration with other global biodiversity data resources.

However, global alignment does not always meet the practical requirements of national-scale conservation and data management. As outlined in Section 2, taxonomic changes can have significant implications for data integrity and conservation outcomes. Regional deviations are now uncommon (e.g. OSME 2025) and this list permits limited, transparent deviations from global taxonomies where necessary.

Key principle: *Global taxonomies provide the defaults for classification, but departures are allowed where required to preserve data integrity, biological resolution, or conservation relevance.*

All deviations from AviList are explicitly documented and tracked (Appendix 1), ensuring that differences are transparent and reversible.

3.1.1. Principles for deviations

Departures from global taxonomies are guided by a small number of explicit principles. These ensure that deviations are applied consistently and only where justified by practical considerations.

1. *Conservation implications*

Where a proposed taxonomic change would reduce the visibility or recognition of a conservation-dependent taxon, a precautionary approach is taken. In such cases, existing classifications may be retained until there is clear and stable justification for change.

Rationale: *Taxonomic revisions do not necessarily reflect changes in extinction risk (see Section 2), but they can affect how taxa are recognised and managed in conservation systems.*

For example, Eastern and Western Hooded Plovers are retained as separate subspecies due to their conservation status and published evidence supporting subspecific separation (Weston *et al.* 2020). AviList has accepted this treatment for a future release.

2. *Loss of data*

Changes that would result in the loss of identifiable biological units, or reduce the resolution at which data can be recorded, may be delayed pending review. This is particularly relevant where distinct forms can be reliably identified in the field or are important for ecological or conservation analyses.

Rationale: *Maintaining the finest practical resolution of biological units (see Section 4) supports long-term data usability and avoids irreversible loss of information.*

For example, Black-browed (*Thalassarche melanophris*) and Campbell (*Thalassarche impavida*) Albatrosses are proposed to be lumped at the species level by AviList (version 1). Black-browed and Campbell Albatrosses are allopatric breeders but have similar non-breeding distributions in Australian waters. The taxa are identifiable at sea. Lumping these species would likely lead to a loss of information, as observers would be prevented from recording an identifiable form (whether a legitimate species, a subspecies, or a non-taxonomic form). Given the distributional overlap in Australian waters, that information would not then be recoverable if a reversion to their current treatment were to occur, which is eminently foreseeable given the history of uncertainty regarding the taxonomic status of these birds.

3. *Distributional uncertainty*

Where proposed taxonomic changes rely on spatial delineations that cannot be clearly or reliably defined, implementation may be deferred. This applies particularly to subspecies whose geographic ranges are poorly described or unresolved.

Rationale: *Spatial data are a core application of this list, and unclear boundaries can compromise mapping, analysis, and interpretation.*

For example, Avilist proposes two subspecies for Chestnut Teal, the distribution of which is continuous through continental Australia. However, there is (at the time of publication) no information available on the biogeographic barriers that may separate the taxa or whether distribution is discrete or whether the subspecies co-occur.

These principles are applied conservatively and transparently. Deviations are:

- explicitly flagged,
- documented with supporting rationale, and
- revisited as new evidence becomes available.

Implementation status (e.g. *implementable*, *not implementable*, *requires checking*) is recorded for each affected taxon (see Appendix 1, Table A1).

Key principle: *Because taxa are linked to stable identifiers (Section 5), alternative classifications can be accommodated without loss of continuity in the underlying data.*

3.2. What constitutes a core Australian bird

This list focuses on **core Australian taxa**, defined as those with ecological or conservation relevance within Australian jurisdictions. A taxon is considered core if it meets one or more of the following criteria (see Table 5 for encoded population values):

- Taxa endemic to Australia.
- Taxa that occur elsewhere, but which have a permanent breeding population(s) within Australia, including small island populations of non-endemic taxa and introduced taxa.
- Taxa that occur elsewhere, for which an ecologically meaningful proportion of their global non-breeding population frequently occurs within Australian territories (nominally $\geq 0.1\%$).

While the third criterion is a policy heuristic, not a strict biological criterion, and estimates may involve speculation, it provides a baseline for consistent judgments until more detailed research in the area elucidates meaningful indices to determine what are core versus vagrant/occasional taxa within a jurisdiction.

** Failed introductions are retained as supplementary listings.*

3.3. Exclusions: vagrant and non-core taxa

This *wlist* includes 'core' Australian birds. Introduced taxa are listed as core taxa, but domestic taxa are not. Vagrant taxa are not given systematic consideration as core taxa are given:

- **Mapping limitations:** The scarcity of records makes meaningful spatial mapping impractical.
- **Data paucity:** insufficient data often exist to accurately estimate Australian population figures, trends, or habitat preferences.
- **Resource intensity:** Significant time is required to ascertain and name subspecies for listing purposes.
- **Extinction risk assessment and conservation status complications:** Their inherently small populations and restricted distributions in Australia can confound conservation assessments against IUCN criteria and potentially lead to irrelevant listings in legislation and treaties.
- **Inherent instability:** the number of taxa recorded as occurring once or a few times in or near Australian territories has grown rapidly in recent decades. This is due, at least in part, to the exponential increase in effort, detection methods, and observer skill, as well as the expansiveness of Australia – i.e., a very large continent with many 'external territories'. The broad extent of external territories brings the nation's political jurisdiction into very remote areas that, in some cases, are more biogeographically within the sphere of other nations. The regularity of occurrence of taxa in outlying regions is therefore difficult to ascertain, and evidence is prone to fluctuations.

Note: *Vagrant taxa can be sourced from global lists such as Avilist Core Team if required.*

In summary...

The scope of this list is defined by a balance between global consistency and national applicability. It:

- adopts global taxonomies as a baseline (Appendix 1),
- applies structured principles to manage necessary deviations, and
- focuses on taxa that are relevant to Australian conservation and ecological data systems.

This approach ensures that the dataset remains both internationally compatible and practically useful for national-scale applications.

4. Biological units

4.1. Species

Species classifications have traditionally been based on a modified Biological Species Concept (Christidis and Boles 2008, del Hoyo *et al.* 2014, del Hoyo *et al.* 2016). Approaches to interpreting evidence vary. BirdLife International used a quantitative framework focused primarily on phenotypic traits to delineate species (Tobias *et al.* 2010). Other lists gave greater weight to genetic evidence. Avilist follows a less prescriptive approach that accommodates genetic information (Avilist Core Team 2025). This list follows Avilist with a small number of exceptions (see Appendix 1).

4.2. Subspecies

Bird subspecies (sometimes also referred to as 'races') are regionally partitioned forms of a species that exhibit significant differences in diagnostic traits (e.g. morphology, colouration or vocalisation). While subspecies are not immutable, they may be synonymised and exhibit clinal artefacts; subspecific trait differences often represent important local adaptations, especially in expansive, environmentally and climatically diverse countries like Australia and the U.S.A. (Winkler 2010).

Unlike species, subspecies with overlapping ranges often interbreed (species do also interbreed, but far less often). This interbreeding, also referred to as hybridisation or introgression, gives rise to 'hybrid zones'.

In these hybrid zones, birds can either breed with other members of their subspecies, or can interbreed with birds from adjacent subspecies. Where interbreeding occurs, 'hybrid forms' exhibiting intermediate characters can arise. Hybrid zones (or zones of introgression) are common in Australia and usually occur along distinct environmental gradients, especially drainage divisions or river basins (Ford 1987b, Ford 1987a, Schodde and Mason 1997, Schodde and Mason 1999). Subspecies with such hybrid zones are known as 'parapatric', while populations that are totally isolated from one another are known as 'allopatric'. This contrasts with the sympatry common to species forms (Tobias *et al.* 2010).

While species are the predominant taxonomic level at which data are recorded, most unique forms of Australian birds are subspecies (Figure 1).

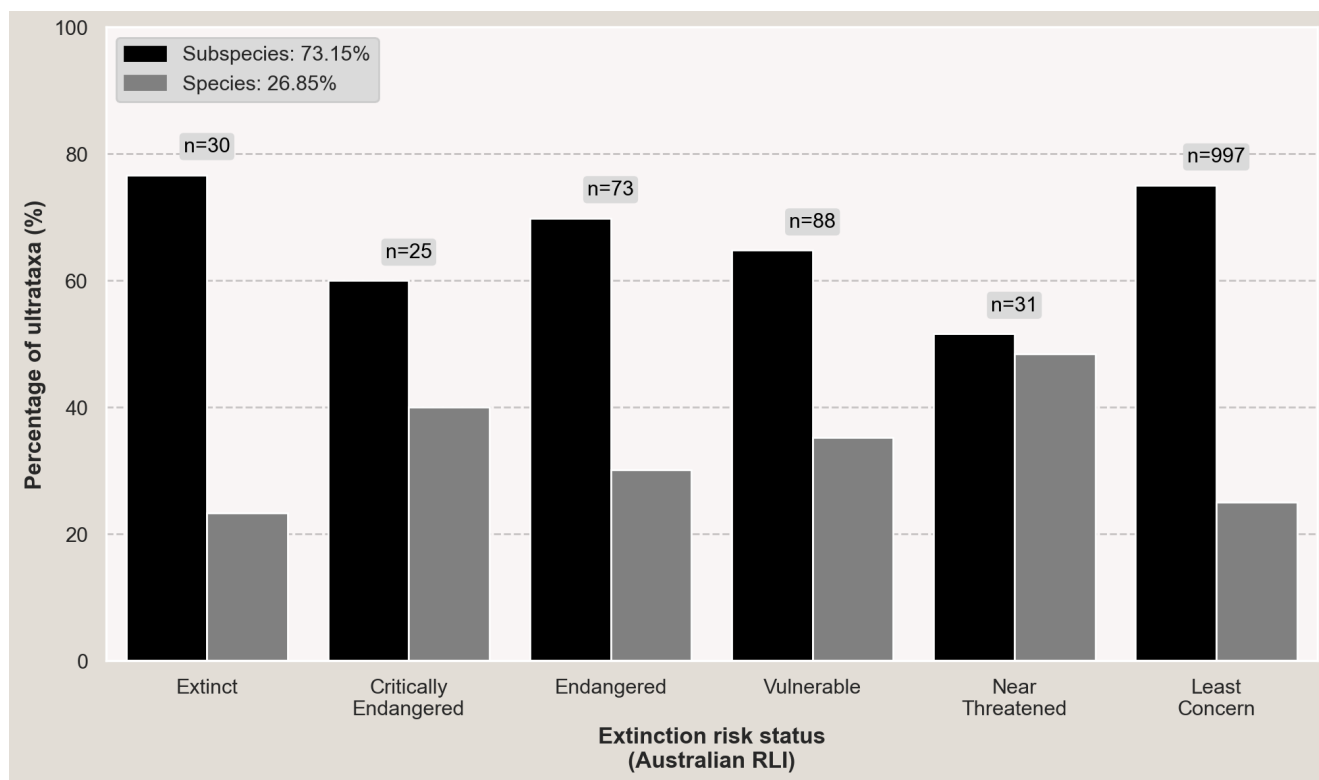


Figure 1: The preponderance of subspecies relative to species ultrataxa. The overall preponderance is shown in the legend.

4.3. Ultrataxa

Terminal taxonomic units (ultrataxa) are a cornerstone of applied conservation research and practice in Australia (e.g. Szabo *et al.* 2012). Developed by Schodde and Mason (1999), an ultrataxon is defined as either a monotypic species (a species with no subspecies) or a subspecies.

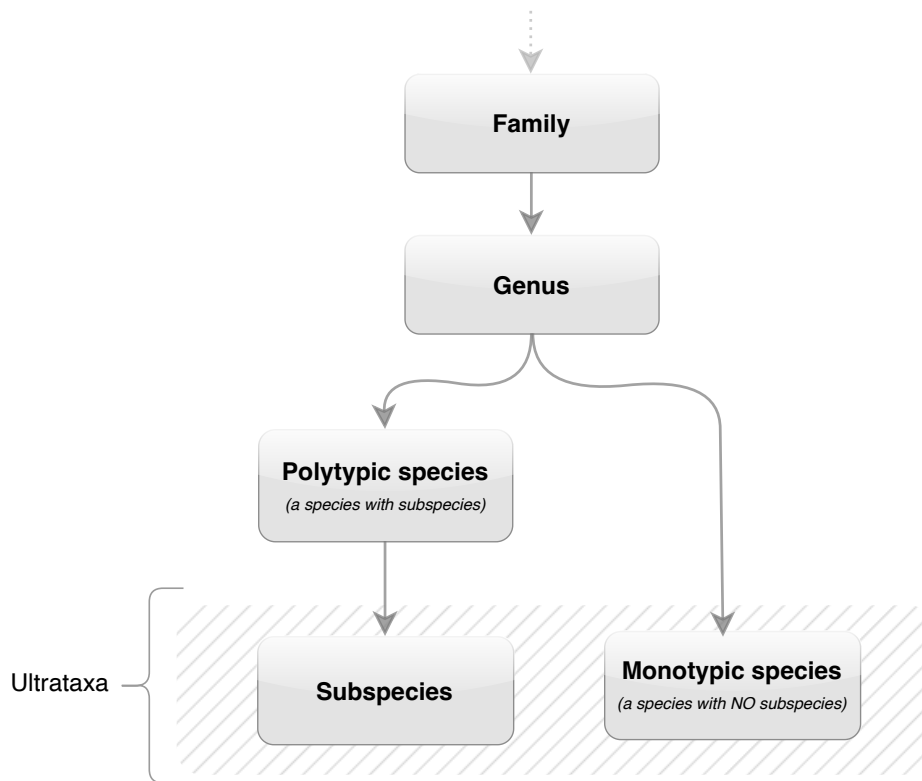


Figure 2: Ultrataxa.

4.4. Higher-order taxonomic classifications

Higher-order taxonomic classifications (those above the level of species) are inconsequential for the management of biodiversity data at the terminal taxonomic level (i.e. ultrataxa). Changes to genera and above can be made to any list without affecting the terminal taxonomic identifiers pertinent to data management and practical conservation of biodiversity. However, a taxonomic sort order is useful, as are high-order classifications for grouping and aggregation of data. Noting the extreme instability of higher-order taxonomic classifications for bird genera and above, the classifications presented here are drawn from the best consistent basis available at the time of publication, and the basis is noted in the general metadata for the dataset.

5. How the list works

5.1. **A working list for spatial and conservation data**

From the comprehensive inventory of species and subspecies, unique and trackable taxonomic identifiers and covariates facilitating the management of taxonomically complex data were developed. A hierarchical system whereby two primary identifiers, comprising a species number (sp_id) and a taxon code (taxon_id), are used to code taxa is employed (Table 2). Values can therefore persist through changes in scientific and common names and link to other data sources, such as spatial layers, monitoring and ecological/traits databases (e.g. Garnett et al. 2015), forming a structural link.

- **sp_id:** is an integer unique to species.
- **taxon_id:** is the unique taxonomic identifier (all rows must have a unique taxon_id). This field is a character concatenation of the sp_id integer, followed by an alphabetical code representing the subspecies (a, b, c, etc.). The ultrataxon identifier ('u') is prefixed to indicate where a taxon is terminal. In this way, the field can be disaggregated into its constituent parts if needed.

Why not just use names as identifiers?

While matching data on names is unavoidable in some cases, stable and persistent unique identifiers are a cornerstone of good data management, ensuring that records remain linkable even when taxonomy changes.

Scientific and common names change over time and may be inconsistently recorded, and text character sets (e.g., non-standard characters and spelling errors) introduce countless permutations of extraneous complexity in a world of already fragmented and disparate datasets that plague data analysts worldwide.

5.2. **Covariates**

Many covariates will be useful for aggregating and filtering spatial range data for applied research. Garnett et al. (2015) have a range of life-history traits and habitat associations that can be linked directly to this data. For this application, however, several additional fields were developed to facilitate systematic decisions on what taxa to map and how those taxa should be handled methodologically.

Bird groups are major categories distinguishing taxa in terms of the methods used to generate range estimates (Table 4). These groupings also reflect fundamental differences in types of avifauna. An additional grouping variable defining obligate and facultative coastal taxa was also derived given the differences required in defining ranges for linearly distributed taxa within the marine grouping. For example, many terns feed primarily (or in some cases wholly) in marine habitats, and hence are marine birds. However, in many cases, terns do not venture far from shorelines, and hence, the geometric unit most closely reflective of distributions is coastlines. Many shorebird distributions also fit this category, although some facultative coastal – i.e. have partial inland distributions.

Bird populations reflect major groupings related to extancy, endemism, breeding status (nationally) and other major grouping variables (Table 5). This variable can be used to exclude non-core populations of birds, such as vagrants, introduced species, domestic populations, and extinct taxa.

6. Technical details

6.1. Data formats

Primary data are available as a PostgreSQL database dump (for full functionality) or comma-separated table data outputs (for general use). Comma-separated value files are exported in the following formats:

- Encoding: UTF-8.
- Quoting: Non-numeric values are quoted; numerics unquoted.
- Delimiter: comma.
- Header = TRUE.

See Table 1 for a complete listing of package files.

Table 1: Data package files

FILE	DESCRIPTION
wlist_readme.pdf	Primary documentation for the package. Covers the conceptual rationale for maintaining a taxonomically stable working inventory of Australian birds, the ultrataxon data model, scope and AviList alignment principles, and full technical implementation including database schema, field definitions, constraints, stored procedures, and the AviList v1 integration summary.
wlist_core.sql	PostgreSQL SQL dump containing CREATE TABLE statements and INSERT data rows for the wlist (core taxon list) and lut_rli (Red List Index lookup) tables. Suitable for direct import into a PostgreSQL database to recreate the working dataset.
wlist.ddl	PostgreSQL Data Definition Language (DDL) script defining the full schema for the wlist and lut_rli tables. Includes primary keys, unique constraints, check constraints (e.g. ultrataxon rules, coastal range codes), foreign key relationships, and indexes. Use this to create the schema without data.
wlist_core.csv	Flat-file CSV export of the core wlist table. Contains 1,739 taxon rows with 20 fields covering taxonomic hierarchy, common and scientific names, Avibase identifiers, bird group, population status, coastal range, and Australian Red List Index (RLI) values for 1990, 2000, 2010, and the current assessment.
lut_rli.csv	Lookup table mapping integer codes to IUCN-aligned Australian Red List Index (RLI) extinction risk categories.
wlist_core_data_dictionary.csv	Field-level data dictionary for the wlist_core table. Documents all 20 fields with name, data type, plain-English description, and (where applicable) a Linked Data IRI linking the field to a controlled vocabulary term in the TERN ontology.
wlist_lut_rli_data_dictionary.csv	Field-level data dictionary for the lut_rli lookup table. Documents the three fields (id, code, category) with name, data type, and plain-English description.
wlist_dqa_report.md	Automated data quality assessment (DQA) report generated against the wlist PostgreSQL database.
avilist_changes.csv	Taxon-level records of taxonomic changes arising from AviList global checklist integration. Each of the 139 rows records a taxon ID, an implementability assessment (Implementable / Not implementable / Partially implementable), and explanatory notes on the nature of the change and any reasons for partial or non-adoption.
wlist_alilist_changes.pdf	PDF companion document to avilist_changes.csv. Provides narrative context, methodology, and detailed explanation of the AviList taxonomy changes assessed for incorporation into the wlist, including the criteria used to determine implementability.
add_taxon/	Folder containing the Python utility for adding rows to wlist via the wlist_add_row procedure. Includes the script (add_taxon.py), a README, and sample CSV and JSON input files. Supports CSV and JSON input, all-or-nothing batch transactions, dry-run validation, and pre-commit checks.
package_wlist.py	Python helper module providing the database connection function (get_db_connection) used by add_taxon. Reads DCORE_HOST, DCORE_DB, DCORE_USER, DCORE_PASSWORD and DCORE_PORT environment variables.

FILE	DESCRIPTION
requirements.txt	Python package dependencies required to run add_taxon (psycpg2-binary).

6.2. Field descriptions

Note: fields below are displayed as in the database schema, which uses 'snake case'. Comma-separated file outputs are encoded in 'camelCase' – for example `taxon_sort` becomes `taxonSort`.

Table 2: Main list fields

FIELD	TYPE	DESCRIPTION
Primary taxonomic fields		
<code>taxon_sort</code>	integer	Sort order for taxa
<code>is_ultrataxon</code>	smallint	1 if this is a terminal taxonomic unit (ultrataxon); otherwise NULL.
<code>taxon_level</code>	character	Label for the taxonomic level. sp = species, - ssp = subspecies
<code>sp_id</code>	smallint	Unique species identifier
<code>taxon_id</code>	character	Unique identifier for each taxon – primary key
<code>taxon_name</code>	character	Taxon common name
<code>taxon_scientific_name</code>	character	Taxon scientific name
<code>family_name</code>	character	Family common name
<code>family_scientific_name</code>	character	Family scientific name
<code>t_order</code>	character	Taxonomic order
<code>avibase_id</code>	character	External Avibase identifier
<code>notes</code>	text	General notes
Covariates		
<code>bird_group</code>	character	Bird group/category – see Table 4
<code>population</code>	character	Population/region qualifier – see Table 5
<code>is_coastal</code>	smallint	Coastal/inland descriptor – exported to csv as character – see Table 6
<code>aust_rli_1990</code>	smallint	Red List Index (1990) – see Table 7
<code>aust_rli_2000</code>	smallint	Red List Index (2000) – see Table 7
<code>aust_rli_2010</code>	smallint	Red List Index (2010) – see Table 7
<code>aust_rli</code>	smallint	Red List Index (current/overall code) – see Table 7
<code>aust_rli_status_desc</code>	character	Description/label matched from <code>lut_rli.category</code>

Table 3: Red list status fields

FIELD	TYPE	DESCRIPTION
id	integer	RLI category identifier (numeric key)
code	character	Short code (e.g., two-letter code)
category	character	Human-readable category/description

6.2.1. Covariate field attributes

Table 4: Spatial bird group attributes

BIRD_GROUP	DESCRIPTION
Terrestrial	Taxa with distributions characterised by irregular occurrence patterns forming polygonal geometries with relatively low edge-interior ratios and which depend on terrestrial habitats (as defined by Garnett et al. 2015).
Wetland	Taxa with distributions characterised by irregular occurrence patterns forming polygonal geometries with relatively low edge-interior ratios and which depend on habitats of inland waters (as defined by Garnett et al. 2015).
Shoreline (migratory/resident)	Taxa associated with linear habitats (shorelines of coasts or inland waters)
Marine	Taxa with distributions characterised by irregular occurrence (point) patterns forming polygonal geometries, and which feed primarily in Marine habitats (as defined by Garnett et al. 2015).

Table 5: Population attributes

POPULATION	DESCRIPTION
Australian	Australian breeding population of taxa that also breed elsewhere
Endemic	Taxa which occur in Australia and nowhere else
Endemic (breeding only)	Taxa which breed only in Australia, but which migrate / disperse frequently elsewhere
Extinct (endemic)	Taxa that occurred naturally only in Australia at the time of European settlement (~1788) but are now extinct
Extinct (Australia only)	Taxa that occurred naturally in Australia at the time of European settlement (~1788) but are now extinct in Australia, but which persist elsewhere
Failed introduction	Taxa introduced to Australia for which wild populations have not persisted (generally by acclimatisation societies early in European settlement).
Introduced	Taxa that have been introduced to Australia since 1788 and established self-sustaining wild populations
Non-breeding	Taxa that occur regularly in Australia but breed elsewhere
Vagrant	Non-breeding taxa that cross into Australian jurisdictional boundaries in insignificant numbers – see section 3 'What constitutes a core Australian bird' and 'Exclusions: vagrants and non-core taxa'
Vagrant (from NZ introduction)	Vagrants derived from populations successfully introduced to New Zealand

* Definitions developed from Garnett et al. (2015).

Table 6: Coastal range attributes

IS_COASTAL	DESCRIPTION
Obligate (1)	Strictly coastal, never found away from coasts
Facultative (2)	Principally coastal, but less frequently found inland

Table 7: Red list status attributes.

VALUE	CODE	DESCRIPTION (RED LIST STATUS)
0	LC	Least Concern
1	NT	Near Threatened
2	VU	Vulnerable
3	EN	Endangered
4	CR	Critically Endangered
5	EX	Extinct

6.3. Data schematics

The schema comprises two tables: a main taxon table (wlist) and a Red List Index lookup table (lut_rli), linked by a foreign key relationship to the master (current) Red List Index field (aust_rli). The historical Red List Index columns in wlist are linked in the same way, associating each taxon's historical and current Red List status with a validated category (Table 7).

Table 8: Table indexes.

INDEX	TABLE	TYPE	COLUMN	PURPOSE
idx_wlist_taxon_id	wlist	UNIQUE	taxon_id	Fast lookup by primary identifier.
idx_wlist_taxon_sort	wlist	UNIQUE	taxon_sort	Fast retrieval in taxonomic sort order.
idx_wlist_sp_id	wlist	Standard	sp_id	Fast lookup for species ID.
wlist_taxon_name_ukey	wlist	Unique	taxon_name	
wlist_taxon_scientific_name_ukey	wlist	Unique	taxon_scientific_name	
idx_wlist_aust_rli	wlist	Standard	aust_rli	Fast filtering by conservation status.
idx_wlist_bird_group	wlist	Standard	bird_group	Fast filtering by ecological group.
idx_lut_rli_id	lut_rli	UNIQUE	id	Fast primary key lookup.
idx_lut_rli_code	lut_rli	Standard	code	Fast lookup by status code.

6.4. Quality control and data modification procedures

See [*wlist_dqa_report.md*](#) (a standard data quality assessment report).

Quality control is maintained through relational database practices, nomenclatural validation, and identifier standardisation. Taxonomic names and classifications are systematically vetted against global lists, with reconciliation via the AviBase identifier providing a persistent crosswalk between taxonomic concepts.

Higher-order taxonomic fields (`family_name`, `family_scientific_name`, `t_order`) are stored as plain text and are not separately normalised, reflecting the known instability of higher-order avian classifications (Table 2). The schema uses CASCADE on update and SET NULL on delete for the foreign key relationship, meaning Red List category updates propagate automatically, while deleting a category will null the associated taxon records rather than blocking the deletion. The `taxon_id` represents the primary stable identifier for each taxon. It is designed to persist through changes in scientific or common names and serves as the linkage key to external datasets such as spatial range layers, trait databases, and occurrence records.

Primary identifiers are designed to be stable and unique, supporting consistent linkage across datasets. All changes are subject to audit logging, validation, and review of updates to taxonomy, nomenclature, and associated attributes (Table 8). Constraints are applied to ensure duplicate values are not propagated, among other functions (Table 9).

Entries are added or deleted via the PostgreSQL procedures `wlist_add_row` and `wlist_delete_row` (Table 10). These procedures work with constraints (Table 9) and triggers (Table 11) to ensure data integrity. The `p_taxon_sort_target` argument is the `taxon_sort` value (row) below which the new entry will be added. Rows (`taxon_sort`) values below this are incremented by the database procedure to maintain taxonomic sequencing. The call examples below highlight correct usage of the data addition/deletion functions. A Python utility (`add_taxon`) provides a batch-friendly, pre-validating wrapper for `wlist_add_row` – see the Programmatic interfaces section that follows.

Table 9: Table constraints.

CONSTRAINT	TABLE	TYPE	APPLIES TO	PURPOSE
wlist_pkey	wlist	PRIMARY KEY	taxon_id	Ensures every taxon has a unique identifier.
wlist_taxon_name_ukey	wlist	UNIQUE	taxon_name	Ensures no two taxa share the same common name.
wlist_taxon_scientific_name_ukey	wlist	UNIQUE	taxon_scientific_name	Ensures no two taxa share the same scientific name.
wlist_sci_name_level_check	wlist	CHECK	taxon_scientific_name	taxon_scientific_name check - ensures species scientific names have 2 words, subspecies have 3.
wlist_aust_rli_fkey	wlist	FOREIGN KEY	aust_rli → lut_rli.id	Ensures Red List values reference valid categories. Cascades on update; sets NULL on delete.
wlist_coastal_range_check	wlist	CHECK	is_coastal	Only accepts valid coastal range codes.
wlist_ssp_ultrataxon_check	wlist	CHECK	is_ultrataxon; taxon_level	Ensures every subspecies (ssp) row is flagged as an ultrataxon; subspecies cannot have is_ultrataxon NULL or FALSE.
wlist_ultrataxon_taxonid_check	wlist	CHECK	taxon_id; is_ultrataxon	Bidirectional ultrataxon check. Prevents taxon_id starting with 'u' when is_ultrataxon is NULL and is_ultrataxon = 1 when taxon_id doesn't start with 'u'
lut_rli_pkey	lut_rli	PRIMARY KEY	id	Ensures each Red List category is uniquely identified.
wlist_aust_rli_fkey	wlist	FOREIGN KEY	aust_rli → lut_rli.id	Ensures Red List values reference valid categories. Cascades on update; sets NULL on delete.
lut_rli_code_ukey	lut_rli	UNIQUE	code	Ensures no duplicate short codes.

Table 10: Functions and procedures

NAME	TYPE	RETURNS	LANGUAGE	PURPOSE
update_is_core	FUNCTION	trigger	plpgsql	Trigger function to set is_core based on population value.
update_ssp_required	FUNCTION	trigger	plpgsql	Trigger function to maintain ssp_required across sibling rows.
update_rli_required	FUNCTION	trigger	plpgsql	Trigger function to maintain rli required flag.
wlist_add_row	PROCEDURE		plpgsql	Procedure to insert a row into wlist and shift taxon_sort values below.
wlist_delete_row	PROCEDURE		plpgsql	Procedure to delete a row from wlist and close the taxon sort gap.

Table 11: Triggers

TRIGGER	EVENTS	LEVEL	FUNCTION	PURPOSE
trg_update_is_core	INSERT OR UPDATE OF population	Row	update_is_core	Maintains is_core flag when population changes or a row is inserted.
trg_update_ssp_required	INSERT OR DELETE OR UPDATE OF is_ultrataxon, taxon_level, sp_id	Row	update_ssp_required	Keeps ssp_required in sync when is_ultrataxon/taxon_level/sp_id change or rows are added/removed.
trg_update_rli_required	INSERT OR UPDATE OF is_core, is_ultrataxon, population	Row	update_rli_required	Flags a core taxon that does not have a current Australian Red List Status assigned.

6.5. Programmatic interfaces

A Python utility, *add_taxon*, provides a higher-level interface to *wlist_add_row*, supporting CSV and JSON input, all-or-nothing transactions, pre-commit validation, and a dry-run mode. Full documentation—including CLI usage, input formats, validation behaviour, and troubleshooting—is provided in *wlist_add_taxon_guide.pdf*, available in the data package.

wlist_add_row inserts a new row into *wlist* below the specified `taxon_sort` position, incrementing subsequent `taxon_sort` values to maintain uniqueness. `sp_id` defaults to the next available value in 5001–7999. `taxon_id` is auto-generated as 'u' || `sp_id` for ultrataxons (`p_is_ultrataxon = 1`). `p_is_ultrataxon` must be explicitly set; NULL will halt execution. Parameter details are in Table 12.

Table 12: *wlist_add_row* parameters

PARAMETER	STATUS	NOTES
<code>p_taxon_sort_target</code>	Required	The <code>taxon_sort</code> value below which the new row will be inserted; subsequent values are incremented.
<code>p_taxon_id</code>	Conditional	Auto-generated as 'u' <code>sp_id</code> for ultrataxons; must be supplied for non-ultrataxons.
<code>p_is_ultrataxon</code>	Required	1 = ultrataxon, 0 = not. NULL will halt execution.
<code>p_taxon_level</code>	Optional	
<code>p_sp_id</code>	Optional	Defaults to the next available value in 5001–7999.
<code>p_taxon_name</code>	Optional	
<code>p_taxon_scientific_name</code>	Optional	
<code>p_family_name</code>	Optional	
<code>p_family_scientific_name</code>	Optional	
<code>p_t_order</code>	Optional	
<code>p_population</code>	Optional	
<code>p_aust_rli_1990</code>	Optional	
<code>p_aust_rli_2000</code>	Optional	
<code>p_aust_rli_2010</code>	Optional	
<code>p_aust_rli</code>	Optional	
<code>p_bird_group</code>	Optional	
<code>p_supplementary</code>	Optional	
<code>p_avibase_id</code>	Optional	
<code>p_reference</code>	Optional	

```

CALL wlist_add_row(
  p_taxon_sort_target      := 81,
  p_taxon_id              := 'u756'::VARCHAR(20),
  p_is_ultrataxon         := 1::SMALLINT,
  p_taxon_level           := 'ssp'::VARCHAR(50),
  p_sp_id                 := 756::SMALLINT,
  p_taxon_name            := 'Lord Howe Metallic Pigeon'::VARCHAR(255),
  p_taxon_scientific_name := 'Columba vitiensis godmanae'::VARCHAR(255),
  p_family_name           := 'Pigeons and Doves'::VARCHAR(255),
  p_family_scientific_name := 'Columbidae'::VARCHAR(255),
  p_t_order               := 'Columbiformes'::VARCHAR(255),
  p_population            := 'Extinct (endemic)'::VARCHAR(255),
  p_bird_group            := 'Terrestrial'::VARCHAR(255),
  p_aust_rli_1990         := 5::SMALLINT,
  p_aust_rli_2000         := 5::SMALLINT,
  p_aust_rli_2010         := 5::SMALLINT,
  p_aust_rli              := 5::SMALLINT
);

CALL wlist_delete_row('u756');

```

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8. Appendices

8.1. **Appendix 1: AviList version 1 integration**

The release of AviList brings 139 putative changes for the core Australian avifauna. Most of these changes are implementable in line with working list principles and scope. Changes not yet implemented in the list are identifiable as having a value other than 'Implementable'. The notes field has the rationales for embellishments. Of the 22 cases where the change is not straightforwardly implemented:

- Nine are 'Not implementable', seven because spatial ranges are not easily definable and the two Hooded Plover subspecies, which must be retained for conservation reasons (AviList will also do so in their 2nd release as indicated in the note).
- Five are 'Partially implementable' (noted as to why).
- Eight require further checking, six of which are seabirds.

All changes – in reference to pre-AviList classifications - are identified and explained in the AviList changes sidecar file '[avilist_changes.csv](#)', which features tracking and implementability status (Table A1). Notes (field; 'description') are written primarily with data change considerations in mind, but potential conservation implications are also highlighted in line with list principles (section 3). The table links to the main list on `taxon_id`. A pdf version '[wlist_avilist_changes.pdf](#)' is also available.

Table A1: *AviList integration fields*

FIELD	TYPE	DESCRIPTION
<code>taxonId</code>	<code>varchar(20)</code>	wlist primary key value.
<code>alistChange</code>	<code>text</code>	Derived label: 'Implementable' 'Not implementable' 'Partially implementable'.
<code>alistChangeNote</code>	<code>text</code>	Free-text note providing additional detail.